

Lecture 3: Linear Least-squares problems

Problem: Given n vectors $a_1, a_2, \dots, a_n \in \mathbb{R}^m$, and a target vector $y \in \mathbb{R}^m$ we want to find coefficients $x_1, x_2, \dots, x_n \in \mathbb{R}$ such that

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = y \quad (\text{or "as close as possible"})$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \quad A = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_n \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{m \times n}$$

$$Ax = y$$

$$\boxed{} = \boxed{}$$

$$\boxed{} = \boxed{}$$

~> for today: this case

$$\boxed{} = \boxed{}$$

Example 1:

Food 1
30 carbs
0 fat
2 proteins

Food 2
40 carbs
50 fat
10 proteins

my ideal diet
40 carbs
20 fat
40 protein

x_1

x_2

$$\begin{bmatrix} 30 \\ 0 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} 40 \\ 50 \\ 10 \end{bmatrix} x_2 = \begin{bmatrix} 40 \\ 20 \\ 40 \end{bmatrix}$$

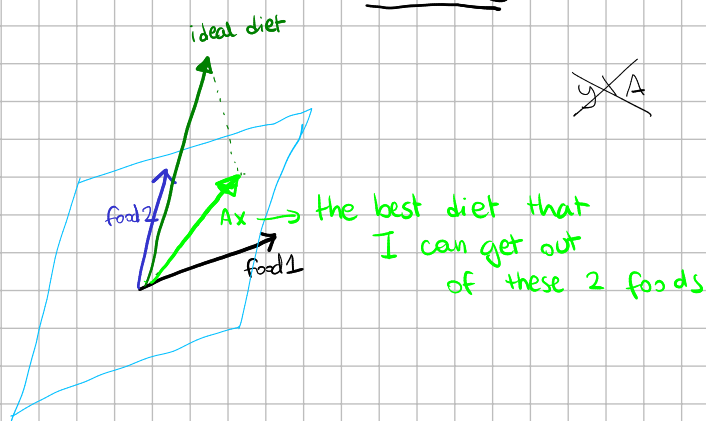
$$\begin{bmatrix} 30 & 40 \\ 0 & 50 \\ 2 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 40 \\ 20 \\ 40 \end{bmatrix}$$

↳ no solution!

$$\underline{x = A \setminus y} \rightsquigarrow \text{solves } \min_x \|Ax - b\|_2$$

~~$y \in A$~~

$y \rightsquigarrow$ column vector



Example 2: NBA salaries

Linear model: $\text{salary} \approx (\text{rebounds}) \cdot x_1 + (\text{fouls}) \cdot x_2 + (\text{points}) \cdot x_3$

Minimize $\sum_p (\text{salary}_p - \text{rebounds}_p \cdot x_1 - \text{fouls}_p \cdot x_2 - \text{points}_p \cdot x_3)^2$ $\otimes$

$$A = \begin{bmatrix} \text{rebounds}_1 & \text{fouls}_1 & \text{points}_1 \\ \text{rebounds}_2 & \text{fouls}_2 & \text{points}_2 \\ \vdots & \vdots & \vdots \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad y = \begin{bmatrix} \text{salary}_1 \\ \text{salary}_2 \\ \vdots \end{bmatrix}$$

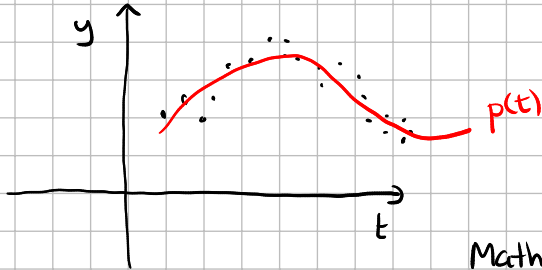
$$\textcircled{*} \Leftrightarrow \min_x \|Ax - b\|_2^2$$

Example 3: polynomial fitting

Given some data points $(t_1, y_1), (t_2, y_2), \dots, (t_m, y_m)$

find a polynomial of degree at most 3 ($p(t)$)

that "best fits" the data



$$p(t) = at^3 + bt^2 + ct + d$$

$$p(t_i) = at_i^3 + bt_i^2 + ct_i + d \text{ for } i=1, \dots, m.$$

↳ want this to be $\approx y_i$

Mathematical formulation:

$$\min_{a,b,c,d} \sum_{i=1}^m (y_i - at_i^3 - bt_i^2 - ct_i - d)^2$$

In matrix form:

$$\min_{a,b,c,d} \left\| \begin{bmatrix} t_1^3 & t_1^2 & t_1 & 1 \\ t_2^3 & t_2^2 & t_2 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ t_m^3 & t_m^2 & t_m & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \right\|_2^2$$

[Useful Matlab trick:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad x.^2 = \begin{bmatrix} x_1^2 \\ x_2^2 \\ x_3^2 \end{bmatrix}$$

$$x, y \text{ vectors } \begin{bmatrix} \end{bmatrix} \begin{bmatrix} \end{bmatrix} \rightarrow x.*y = \begin{bmatrix} x_1 y_1 \\ x_2 y_2 \\ x_3 y_3 \end{bmatrix}$$

also with matrices

Putting $\cdot$ before operations means that

they are done elementwise]

$$\min_x \|Ax - y\|_2$$

$$\min_w \|Xw - y\|_2$$

$$\min_{w_1, \dots, w_n} \left\| \begin{bmatrix} w_1 & \dots & w_n \end{bmatrix} \begin{bmatrix} x \\ \vdots \end{bmatrix} - \begin{bmatrix} y \\ \vdots \end{bmatrix} \right\|_2$$

Uniqueness of solution (?)

Example: $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $y = \begin{bmatrix} 4 \\ 4 \\ 1 \\ 2 \end{bmatrix}$ $Ax = y$

$\leadsto$ closest vector we can get is $\begin{bmatrix} 4 \\ 4 \\ 0 \\ 0 \end{bmatrix}$

$x = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, $x = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$, ...

Recall: $\text{Ker}(A) = \{z \text{ s.t. } Az = 0\}$ $\rightarrow$ in this case, solution is not unique!

$$\text{ker}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \right\}$$

what's the problem? if $\exists z \neq 0$ s.t. $Az = 0$

if x is a solution of the least-squares problem, then

$$\text{also } x+z \text{ is a solution: } A \cdot (x+z) = Ax + Az = Ax$$

Def: The matrix $A \in \mathbb{R}^{m \times n}$ has **FULL COLUMN RANK** if $\text{Ker}(A) = \{0\}$

Equivalently, if the rank of A is the number of columns

Equivalently, if there is no vector $z \neq 0$ such that $Az = 0$.



Lemma: $A \in \mathbb{R}^{m \times n}$ has full column rank $\Leftrightarrow A^T A$ is SPD

Proof: A has full column rank $\Leftrightarrow Az \neq 0 \ \forall z \neq 0$

$$\Leftrightarrow \|Az\|^2 \neq 0 \ \forall z \neq 0 \Leftrightarrow \|Az\|^2 > 0 \ \forall z \neq 0$$

$$\Leftrightarrow (Az)^T (Az) > 0 \ \forall z \neq 0 \Leftrightarrow z^T A^T A z > 0 \ \forall z \neq 0 \Leftrightarrow A^T A \text{ SPD. } \square$$

$\min_x \|Ax - y\|_2^2 \leadsto$ we want to rewrite it as a quadratic optimization problem

$$\|Ax - y\|^2 = (Ax - y)^T (Ax - y) = (x^T A^T - y^T) (Ax - y)$$

$$= x^T A^T A x - \underbrace{x^T A^T y}_{\langle A^T y, x \rangle} - \underbrace{y^T A x}_{\langle x, A^T y \rangle} + y^T y$$

$$= x^T A^T A x - 2 \langle A^T y, x \rangle + y^T y$$

$$\min_x \frac{1}{2} \|Ax - y\|^2 = \min_x \left(\frac{1}{2} \underbrace{x^T A^T A x}_Q - \langle A^T y, x \rangle + \frac{1}{2} \langle y, y \rangle \right)$$

Letting $Q = A^T A$, $q = -A^T y$,

$$\min_x \frac{1}{2} x^T Q x + \langle q, x \rangle + \frac{1}{2} \langle y, y \rangle$$

↳ this has a unique solution when Q is SPD

↳ gradient of this is $Qx + q = A^T A x - A^T y$

gradient of our objective function is 0 $\Leftrightarrow A^T A x - A^T y = 0$

Theorem: The least-squares problem $\min_x \|Ax - b\|$ has a unique solution $\Leftrightarrow A$ has full column rank.

$$A \in \mathbb{R}^{m \times n}$$

Moreover, we get a strategy to solve a least-squares problem:

1. Compute $A^T A$ $n \times m \times m \rightsquigarrow O(mn^2)$
2. Compute $A^T y$ $m \times n \times m \rightsquigarrow O(mn)$
3. Solve $(A^T A)x = A^T y \rightsquigarrow O(n^3)$

Some tricks to speed up a bit the computations:

$$\begin{bmatrix} A^T \end{bmatrix} \begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} \text{upper triangular} \end{bmatrix} \rightarrow \text{only need to compute the upper triangular part, because } A^T A \text{ is symmetric}$$

$$[A^T A] x = A^T y$$

↳ Cholesky decomposition $A^T A = LL^T = \begin{bmatrix} \text{upper triangular} \end{bmatrix} \begin{bmatrix} \text{upper triangular} \end{bmatrix}$

→ speed up by a factor ≈ 2 .

$A^T A x = A^T y$ "normal equations"

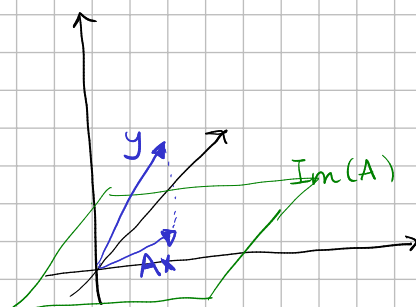
$$\hookrightarrow x = (A^T A)^{-1} A^T y$$

$$Ax = A(A^T A)^{-1} A^T y$$

↳ "reconstructed" vector

seen as a linear transformation, this is the orthogonal projection onto $\text{Im}(A) = \text{span of the columns of } A$.

this is why they are called "normal" equations.



Def: For $A \in \mathbb{R}^{m \times n}$ with full column rank, we define

$A^+ = (A^T A)^{-1} A^T$ to be the PSEUDOINVERSE of A

↳ ("Moore-Penrose inverse")

↳ the solution of $\min_x \|Ax - y\|$ is $x = A^+ \cdot y$

Properties:

- $A^+ \cdot A = (A^T A)^{-1} A^T A = I$

- if A is square $n \times n$ and

invertible, then $A^+ = (A^T A)^{-1} A^T = A^{-1} \underbrace{A^{-T} \cdot A^T}_I = A^{-1}$

→ pseudoinverse generalizes the concept of

"inverse" for rectangular matrices

- $A \cdot A^+ = \underbrace{\begin{bmatrix} \end{bmatrix}}_m \cdot \underbrace{\begin{bmatrix} \end{bmatrix}}_n \boxed{\neq I}$

Remark:

$$A = \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}_{100 \times 5} \quad A^+ = \begin{bmatrix} & & & & \end{bmatrix}_{100 \times 5}$$